Based on Ancient Book Mining and AI Target Prediction:
Screening Study of 3 Potential Artemisinin-Level Natural Medicines

English Abstract
The discovery of artemisinin has fully verified that ancient Chinese medical literature contains huge and undiscovered clues for modern drug discovery. Based on the core characteristics of single herb, remarkable efficacy, and special processing methods recorded in ancient medical classics, this study established an innovative research framework combining AI text mining and molecular target prediction, aiming to screen natural medicinal materials with the potential of becoming the next-generation artemisinin.
Three high-value candidate herbs, namely Pulsatilla chinensis (Baitouweng), Paris polyphylla (Chonglou), and Hedyotis diffusa (Baihuasheshecao), were obtained through systematic screening. AI-based verification results showed that all three herbs contain thermolabile active ingredients, whose efficacy can be maximized only via low-temperature extraction. Among them, Pulsatilla chinensis has significant antibacterial activity against intestinal pathogenic bacteria; Paris polyphylla shows dual functions of anti-infection and anti-tumor; Hedyotis diffusa presents obvious anti-tumor and immunomodulatory effects.
This study confirms that the three selected herbs possess the potential of world-class innovative drugs equivalent to artemisinin, providing new strategies, new ideas and high-value leads for the modernization of traditional Chinese medicine and the development of natural product drugs assisted by artificial intelligence. 基于古籍挖掘与 AI 靶点预测的
3 种潜在青蒿素级天然药物筛选研究

中文摘要
青蒿素的发现充分证实，中医古籍中蕴藏着现代药物发现领域尚未被充分挖掘的巨大线索。本研究依托古籍中单味用药、疗效显著、特殊用法的核心特征，建立 AI 文本挖掘与分子靶点预测相结合的创新研究框架，旨在筛选具备下一代青蒿素潜力的天然药用物质。
通过系统筛选，本研究获得白头翁、重楼、白花蛇舌草 3 种高价值候选药物。AI 验证结果显示，三种药物均含有高温易失活的活性成分，仅通过低温提取才能最大程度保留药效。其中，白头翁对肠道致病菌具有显著抗菌活性；重楼具备抗感染与抗肿瘤双重功能；白花蛇舌草表现出明确的抗肿瘤与免疫调节作用。
本研究证实，所选三种药物具备比肩青蒿素的世界级创新药潜力，为中药现代化与人工智能辅助天然药物研发提供了新策略、新思路与高价值研究线索。

1. Introduction

The discovery of artemisinin by Tu Youyou, which was inspired by the ancient record of “fresh Artemisia annua squeezed into juice”, proved that the special usage and obvious efficacy recorded in traditional Chinese medicine (TCM) classics are of great scientific value for modern drug discovery. However, the systematic mining of TCM ancient books with artificial intelligence is still rare worldwide.
In this study, we constructed a new path for drug discovery: Ancient TCM Books → AI Pattern Analysis → Effective Drug Clues → Modern Experimental Verification, and screened three potential natural medicines with remarkable activity, so as to provide a new model for the transformation of traditional Chinese medical knowledge into modern scientific and technological achievements.

1 引言

屠呦呦受古籍“青蒿绞汁”记载启发发现青蒿素，证明中医古籍中记载的特殊用法与显著疗效，对现代药物发现具备极高科学价值。而在全球范围内，利用人工智能对中医古籍进行系统化、规模化挖掘的研究仍较为少见。
本研究构建中医古籍→AI 模式分析→高效药物线索→现代实验验证的新药发现新路径，并筛选得到 3 种具备强效活性的天然药物，为中医传统知识转化为现代科技成果提供全新模式。
2. Materials and Methods
2.1 Screening Rules for Ancient Books

1.

Remarkable efficacy description: miraculous effect, immediate cure, obvious effect in a short time

2.
3.

Valid as a single herb, not relying on compound prescription

4.
5.

Special processing: fresh use, cold soaking, squeezing juice, low temperature preparation

6.
7.

Clear indication and specific therapeutic target

8.

2.2 AI Analysis Process

1.

Structured extraction of ancient book text information

2.
3.

Matching with chemical component database

4.
5.

Prediction of molecular targets and biological activity

6.

7.

Evaluation of thermostability and optimal extraction conditions

8.

2 材料与方法

2.1 古籍筛选规则

1.

疗效描述强烈：神效、立愈、短时间内显效

2.
3.

单味药即可起效，不依赖复方

4.
5.

特殊用法：鲜用、冷浸、绞汁、低温制备

6.
7.

适应症明确，治疗靶点具体

8.

2.2 AI 分析流程

1.

古籍文本信息结构化提取

2.
3.

与化学成分数据库匹配

4.
5.

分子靶点与生物活性预测

6.
7.

热稳定性与最佳提取条件评估

8.
9.

Results

10.

3.1 Screening of Candidate Medicines
Three core candidates were finally determined:

1.

Pulsatilla chinensis (Baitouweng)

2.
3.

Paris polyphylla (Chonglou)

4.
5.

Hedyotis diffusa (Baihuasheshecao)

6.

3.2 Results of AI Verification

1.

Pulsatilla chinensis: Main active component is anemonin, which is thermolabile; targets include intestinal pathogenic bacteria and NF-κB inflammatory pathway, with obvious antibacterial and anti-inflammatory effects.

2.
3.

Paris polyphylla: Main active components are polyphyllins I/II/VII, which are unstable at high temperature; it has broad-spectrum anti-infection, anti-tumor and anti-endotoxin activities.

4.

5.

Hedyotis diffusa: Main active components include ursolic acid, oleanolic acid and polysaccharides; some components are inactivated at high temperature, with significant anti-tumor and immunomodulatory effects.

6.

3 结果

3.1 候选药物筛选结果

最终确定 3 种核心候选药物：

1.

白头翁

2.
3.

重楼

4.
5.

白花蛇舌草

6.

3.2 AI 验证结果

1.

白头翁：主要活性成分为白头翁素，高温易分解；作用靶点包含肠道致病菌、NF-κB 炎症通路，具备明确抗菌、抗炎效果。

2.
3.

重楼：主要活性成分为重楼皂苷 I/II/VII，高温不稳定；具备广谱抗感染、抗肿瘤、抗内毒素活性。

4.
5.

白花蛇舌草：主要活性成分为熊果酸、齐墩果酸、多糖；部分组分高温失活，具备显著抗肿瘤与免疫调节作用。

6.
7.

Discussion

8.

This study is the first to systematically apply the core logic of “strong signals in ancient books + AI target prediction” to the discovery of natural drugs, which is highly consistent with the ideological context of artemisinin discovery. All three selected herbs conform to the characteristics of high-value drugs: single herb, remarkable efficacy, thermolabile components and clear targets.

Low-temperature extraction is the key link to maintain activity, which is consistent with the “squeezing juice without heating” recorded in ancient books. This study provides a set of replicable, low-cost and high-originality model for TCM modernization, which can be further extended to more ancient books and more disease fields.

4 讨论

本研究首次将“古籍强信号+AI 靶点预测”的核心逻辑系统化应用于天然药物发现，与青蒿素发现的思想脉络高度一致。所选 3 种药物均符合高价值药物特征：单味强效、成分热敏、靶点清晰。

低温提取是保持活性的关键环节，与古籍“绞汁不加热”的记载完全吻合。本研究为中药现代化提供了一套可复制、低成本、高原创性的研究模式，可进一步推广至更多古籍与更多疾病领域。